

Below I'll give you a fully *real-analysis-only* derivation of the small- x behavior of the Fabius function $F(x)$.

1) Exact identity on a dyadic logarithmic scale

Set

$$t := \log_2 \frac{1}{x} = \frac{\ln(1/x)}{\ln 2}, \quad \phi(t) := F(2^{-t}), \quad g(t) := -\ln \phi(t) = -\ln F(2^{-t}).$$

We compute $\phi'(t)$ by the chain rule:

$$\phi'(t) = \frac{d}{dt} F(2^{-t}) = F'(2^{-t}) \cdot \frac{d}{dt} (2^{-t}) = F'(2^{-t}) \cdot (-\ln 2) 2^{-t}.$$

Using $F'(x) = 2F(2x)$ at $x = 2^{-t}$ gives $F'(2^{-t}) = 2F(2^{1-t})$, hence

$$\phi'(t) = -(\ln 2) 2^{1-t} F(2^{1-t}) = -(\ln 2) 2^{1-t} \phi(t-1).$$

Since $\phi(t) > 0$ and $\phi'(t) < 0$, both $\ln \phi$ and $\ln(-\phi')$ are defined. Rearranging and taking logs,

$$\ln \phi(t-1) - \ln \phi(t) = \ln(-\phi'(t)) - \ln(\ln 2) - (1-t) \ln 2 - \ln \phi(t).$$

With $g = -\ln \phi$ and $g'(t) = -\phi'(t)/\phi(t) > 0$, this becomes the exact identity

$$\boxed{g(t) - g(t-1) = \ln g'(t) - \ln(\ln 2) - (1-t) \ln 2.} \quad (1)$$

2) Large- t expansion of g : determination of the coefficients

All $O(\cdot)$ estimates below are taken as $t \rightarrow \infty$ (that is, $x \rightarrow 0^+$).

We will use the standard asymptotic expansions

$$\ln(t-1) = \ln t - \frac{1}{t} - \frac{1}{2t^2} + O\left(\frac{1}{t^3}\right), \quad \ln \frac{t}{t-1} = \frac{1}{t} + \frac{1}{2t^2} + O\left(\frac{1}{t^3}\right). \quad (2)$$

Seek g of the form

$$g(t) = \frac{\ln 2}{2} t^2 + A t \ln t + B t + P (\ln t)^2 + C \ln t + R(t), \quad (3)$$

with constants A, B, P, C to be determined and a remainder $R(t)$ to be bounded later.

2.1 Quadratic coefficient

If $g(t) \sim \alpha t^2/2$, then $g(t) - g(t-1) \sim \alpha t$ and $\ln g'(t) = \ln(\alpha t) = \ln t + \ln \alpha$. Comparing the t -terms in (1), whose right-hand side contains $(t-1) \ln 2 = t \ln 2 - \ln 2$, forces $\alpha = \ln 2$. Thus the quadratic coefficient in (3) is $\ln 2/2$ as written.

2.2 The $t \ln t$ and t coefficients

Compute $g(t) - g(t-1)$ from (3), using (2). Each summand contributes:

- Quadratic part:

$$\frac{\ln 2}{2} (t^2 - (t-1)^2) = \ln 2 t - \frac{\ln 2}{2}.$$

- $A t \ln t$ part:

$$A [t \ln t - (t-1) \ln(t-1)] = A \left(\ln t + 1 - \frac{1}{2t} - \frac{1}{6t^2} \right) + O\left(\frac{1}{t^3}\right).$$

- $B t$ part: $= B$.
- $P (\ln t)^2$ part (use $\ln t - \ln(t-1) = \frac{1}{t} + \frac{1}{2t^2} + O(\frac{1}{t^3})$):

$$P [(\ln t)^2 - (\ln(t-1))^2] = 2P \frac{\ln t}{t} + P \frac{\ln t - 1}{t^2} + O\left(\frac{\ln t}{t^3}\right).$$

- $C \ln t$ part:

$$C [\ln t - \ln(t-1)] = C \left(\frac{1}{t} + \frac{1}{2t^2} \right) + O\left(\frac{1}{t^3}\right).$$

Adding,

$$\begin{aligned}
g(t) - g(t-1) &= \ln 2t - \frac{\ln 2}{2} + A \ln t + A + B \\
&\quad + \frac{-\frac{A}{2} + C + 2P \ln t}{t} + \frac{-\frac{A}{6} + \frac{C}{2} + P \ln t - P}{t^2} + O\left(\frac{\ln t}{t^3}\right).
\end{aligned} \tag{4}$$

Next expand $\ln g'(t)$. Differentiate (3):

$$g'(t) = \ln 2t + A(\ln t + 1) + B + \frac{2P \ln t}{t} + \frac{C}{t} + R'(t).$$

For $\ln(u + \delta)$ we use

$$\ln(u + \delta) = \ln u + \frac{\delta}{u} - \frac{\delta^2}{2u^2} + O\left(\frac{|\delta|^3}{|u|^3}\right) \quad \text{when } |\delta| \leq \frac{1}{2}|u|. \tag{5}$$

Here $u = \ln 2t$, and for large t the ratio $|\delta|/|u| = O(\frac{\ln t}{t})$, so (5) applies. A direct substitution gives

$$\begin{aligned}
\ln g'(t) &= \ln(\ln 2) + \ln t + \frac{A \ln t + A + B}{\ln 2} \cdot \frac{1}{t} \\
&\quad + \left(\frac{C + 2P \ln t}{\ln 2} - \frac{(A \ln t + A + B)^2}{2(\ln 2)^2} \right) \frac{1}{t^2} + O\left(\frac{(\ln t)^3}{t^3}\right).
\end{aligned} \tag{6}$$

Subtracting $\ln(\ln 2) + (1-t) \ln 2$ (the remaining terms of the right-hand side of (1)) yields

$$\begin{aligned}
\text{RHS of (1)} &= \ln 2t - \ln 2 + \ln t \\
&\quad + \frac{A \ln t + A + B}{\ln 2} \cdot \frac{1}{t} \\
&\quad + \left(\frac{C + 2P \ln t}{\ln 2} - \frac{(A \ln t + A + B)^2}{2(\ln 2)^2} \right) \frac{1}{t^2} + O\left(\frac{(\ln t)^3}{t^3}\right).
\end{aligned} \tag{7}$$

Match (4) with (7):

- Coefficient of $\ln t$ (constant level): $A = 1$.
- Constant term: $-\frac{\ln 2}{2} + A + B = -\ln 2$ with $A = 1$, hence

$$B = -1 - \frac{\ln 2}{2}.$$

2.3 The $(\ln t)^2$ and $\ln t$ coefficients

Match the $1/t$ terms in (4) and (7) using $A = 1$ and the above B :

$$\frac{-\frac{1}{2} + C + 2P \ln t}{t} = \frac{\ln t - \frac{\ln 2}{2}}{\ln 2} \cdot \frac{1}{t}.$$

Therefore

$$\boxed{P = \frac{1}{2 \ln 2}, \quad C = 0.}$$

With these coefficients,

$$\boxed{g(t) = \frac{\ln 2}{2} t^2 + t \ln t - \left(1 + \frac{\ln 2}{2}\right)t + \frac{(\ln t)^2}{2 \ln 2} + R(t).} \quad (8)$$

3) Remainder and the 1-periodic term

Define $G(t)$ to be the explicit part of $g(t)$ in (8), and set

$$H(t) := g(t) - G(t) = R(t).$$

Subtract (1) written for G from (1) written for g :

$$H(t) - H(t-1) = [\ln g'(t) - \ln G'(t)] + \rho(t), \quad (9)$$

where $\rho(t) = O((\ln t)^2/t^2)$ collects the first unmatched terms of order $1/t^2$ from (4)–(7).

Since

$$G'(t) = \ln 2 t + (\ln t) + O(1) \quad (t \rightarrow \infty),$$

we have $G'(t) \asymp t$. Write $g'(t) = G'(t) + H'(t)$ and expand

$$\ln g'(t) - \ln G'(t) = \ln \left(1 + \frac{H'(t)}{G'(t)}\right) = \frac{H'(t)}{G'(t)} + O\left(\frac{H'(t)^2}{t^2}\right).$$

Hence (9) becomes

$$H(t) - H(t-1) = \frac{H'(t)}{\ln 2 t} \left(1 + O\left(\frac{\ln t}{t}\right)\right) + O\left(\frac{1}{t^2}\right). \quad (10)$$

Fix $\theta \in [0, 1)$ and put $t_n := n + \theta$. Summing (10) from $n = 1$ to N gives

$$H(t_N) - H(t_0) = \sum_{n=1}^N \frac{H'(t_n)}{\ln 2 t_n} + O\left(\sum_{n=1}^N \frac{\ln t_n}{t_n^2}\right).$$

The second sum is $O(1)$. Using summation by parts on the first sum and $G'(t) \asymp t$ shows that its tails are $O(\frac{\ln N}{N})$. Therefore the limit

$$\Psi(\theta) := \lim_{N \rightarrow \infty} \left(H(t_N) - \sum_{n=1}^N \frac{H'(t_n)}{\ln 2 t_n} \right)$$

exists, is bounded in θ , and satisfies $\Psi(\theta + 1) = \Psi(\theta)$. Writing H as $\Psi(\{t\})$ plus the tail of the two series yields

$$H(t) = \Psi(\{t\}) + E(t), \quad E(t) = O\left(\frac{\ln t}{t}\right), \quad (11)$$

where $\{t\} = t - \lfloor t \rfloor$ is the fractional part of t . This proves the required decomposition of the remainder.

4) Final formula for $\ln F(x)$ as $x \rightarrow 0^+$

Recall $g(t) = -\ln F(2^{-t})$ and $t = \log_2(1/x)$. Negating (8)–(11):

$$\begin{aligned} \ln F(x) = & -\frac{\ln 2}{2} t^2 - t \ln t - \frac{(\ln t)^2}{2 \ln 2} + \left(1 + \frac{\ln 2}{2}\right) t \\ & + \Psi(\{t\}) + O\left(\frac{\ln t}{t}\right), \quad t = \log_2 \frac{1}{x} \rightarrow \infty. \end{aligned}$$

Here Ψ is a bounded 1-periodic function determined by the normalization of F (rescaling F by a constant adds a constant to Ψ). The remainder $O(\frac{\ln t}{t})$ is strictly smaller than any displayed term.